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Unit II: Plant Morphology and Taxonomy of Angiosperm

Chapter 3

Vegetative Morphology

Learning Objectives

The learner will be able to,

- *Explore the parts of the flowering plants*
- *Differentiate vegetative morphology and reproductive morphology*
- *Compare various root systems and their modifications*
- *Understand the stem modifications and functions*
- *Interpret the structure of leaf and functions of leaf*

Chapter Outline



3.1 Habit
3.2 Plant habitat
3.3 Life Span
3.4 Parts of a flowering plant
3.5 Root System
3.6 Shoot system
3.7 Leaf

The study of various external features of the organism is known as **morphology. Plant morphology** also known as **external morphology** deals with the study of shape, size and structure of plants and their parts (roots, stems, leaves, flowers, fruits and seeds). Study of morphology is important in taxonomy. Morphological features are important in determining productivity of crops. Morphological characters indicate the specific habitats of living as well as the fossil plants and help to correlate the distribution in space and time of fossil plants. Morphological features are also significant for phylogeny.

Plant Morphology can be studied under two broad categories:

A. **Vegetative morphology** – It includes shoot system and root system
B. **Reproductive morphology** – It includes Flower/inflorescence, Fruit and Seed

A. Vegetative morphology

Vegetative morphology deals with the study of shape, size and structure of plants and their parts roots, stems and leaves. To understand the vegetative morphology the following important components are to be studied. They are, 1) Habit, 2) Habitat and 3) Lifespan.